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OrbitReady: A Source-Linked Pre-Review Copilot for Spacecraft Debris-Mitigation Evidence

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Abstract

Small-spacecraft teams must assemble mission parameters, disposal evidence, passivation evidence, and jurisdiction-specific guidance before expert or regulatory review. These facts are often distributed across mechanical drawings, propulsion data sheets, and spreadsheets. We present **OrbitReady** (incorporating the OrbitScore evidence-prioritization heuristics), a prototype source-linked pre-review copilot that extracts or accepts mission facts, applies versioned deterministic rule fixtures, identifies missing evidence, compares a controlled counterfactual, and exports a cited review packet. The prototype separates document assistance from rule evaluation and requires qualified human review for critical findings. Demonstrations use labeled synthetic mission profiles and public guidance. OrbitReady is not an official regulator, certification tool, orbital collision-prediction system, operator-facing conjunction-alert service, or replacement for NASA Debris Assessment Software (DAS). The prototype's readiness index is an engineering prioritization heuristic, not an empirical safety probability.

1 Introduction & The Orbital Congestion Challenge

The operational environment of Low Earth Orbit (LEO, defined as the orbital shell below 2,000 km altitude) is experiencing an unprecedented structural transformation. Over the past five years, the operational satellite population has increased substantially, driven primarily by commercial constellations deployed in dense orbital shells between 500 km and 1,200 km. Concurrently, the tracked catalog maintained by the U.S. Space Force 18th Space Defense Squadron (18 SDS) encompasses over 36,000 debris objects larger than 10 cm, while statistical models developed by NASA and the European Space Agency (ESA) estimate a large background population of untracked fragments that pose impact hazards to operational space systems.

The theoretical foundation of orbital runaway collision cascades was first formalized by Kessler and Cour-Palais (1978), demonstrating that as the spatial density of satellites increases, random hypervelocity mutual collisions generate fragments at a rate exceeding atmospheric drag removal, eventually creating an exponential, self-sustaining debris belt [1]. In a subsequent treatise, Kessler (1991) investigated collisional cascading and population growth limits in LEO, showing that dense altitude bands face critical long-term population thresholds [2]. Modern empirical modeling by Liou and Johnson (2006) utilized NASA's LEGEND evolutionary model to examine long-term debris-environment risk and evaluate the modeled effectiveness of various mitigation scenarios [3].

To address near-Earth orbital sustainability, international regulatory bodies and space agencies publish distinct debris mitigation rules and guidance. Most notably, the U.S. Federal Communications Commission (FCC) adopted Report and Order 22-74 in September 2022, legally mandating that space stations ending their mission in or passing through LEO below 2,000 km must complete disposal via atmospheric reentry within five years when planning uncontrolled atmospheric disposal, within the FCC's licensing and U.S. market-access scope [4]. In parallel, NASA NPR

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8715.6E and NASA Technical Standard NASA-STD-8719.14C specify program and project requirements for NASA missions, including accidental explosion passivation (Req 4.4-2), collision with large objects (Req 4.5-1: $P_{coll} < 0.001$), post-mission disposal reliability (Req 4.6-4: $P_{pmd} \geq 0.90$), and uncontrolled atmospheric reentry human casualty risk (Req 4.7-1: $E_c < 10^{-4}$, alongside a 15 J kinetic energy threshold for potentially hazardous debris fragments) [5, 6]. Across Europe, ESA's Space Debris Mitigation Requirements (ESSB-ST-U-007) and ISO 24113:2023 define project and consensus standards, while ISRO's initiative document outlines India's 2030 sustainability pledge [7, 8, 39].

Core Principle & Safety Boundary: OrbitReady is a source-linked pre-review and evidence-prioritization prototype. It is not an official certification authority, regulatory body, NASA DAS replacement, orbital collision prediction system, or operational conjunction alert tool. It is designed solely as an engineering pre-screening assistant.

2 Problem Archaeology: The Operator Bottleneck

While global space agencies and well-funded constellation prime contractors maintain dedicated departments for regulatory affairs and orbital safety, emerging small satellite operators, commercial NewSpace startups, and university CubeSat teams face an acute pre-review bottleneck. These operators represent a rapidly growing segment of LEO launches, yet they are structurally ill-equipped to navigate complex, multi-jurisdictional compliance evidence workflows.

2.1 Current Workflow & Tooling Fragmentation

Today, preparing an Orbital Debris Assessment Report (ODAR) or an End-of-Mission Plan (EOMP) requires operators to manually extract parameters from mechanical CAD drawings, propulsion data sheets, and orbital ephemerides, manually inputting these values into desktop software tools such as NASA's Debris Assessment Software (DAS 3.2) or ESA's Debris Risk Assessment and Mitigation Analysis (DRAMA 3.0) [9, 10]. Both tools serve as specialized desktop calculation suites rather than modern API-integrated CI/CD pre-screening assistants. Meanwhile, commercial SSA platforms focus on live on-orbit tracking and space traffic management rather than pre-licensing evidence assembly [11, 12].

2.2 Anatomy of Pre-Review Evidence Gaps

Our field review of small satellite licensing filings identifies four recurrent evidence gaps: (1) Lifetime Calculation Ambiguities, where inaccurate ballistic coefficient assumptions lead to post-mission lifetimes exceeding statutory targets; (2) Passivation Deficits, failing to document depletion, venting, or isolation plans that prevent qualifying debris-producing explosions under NASA Req 4.4-2 [13, 14]; (3) Conjunction Screening Gaps in crowded orbital regimes; and (4) Reentry Casualty Area Uncertainty, neglecting aerothermal demise analysis for high-melting-point materials that survive atmospheric reentry.

3 Statutory Guidance & Standards Synthesis

To ensure technical precision and prevent jurisdictional overreach, OrbitReady synthesizes primary debris-mitigation standards into versioned rule profiles. Table 1 summarizes the primary statutory mandates and explicit engineering limits referenced within the OrbitReady evaluation fixtures.

Table 1: Debris mitigation statutory standards, agency requirements, and voluntary guidance referenced by OrbitReady.

Authority & Standard	Specific Scope & Quantitative Engineering Threshold
FCC R&O; 22-74 (2022)	5-Year disposal rule for LEO satellites (<2,000 km) planning uncontrolled atmospheric reentry; FCC licensing & U.S. market access scope.
NASA-STD-8719.14C Req 4.3	Debris released during normal operations: Object-time product < 100 object-years for fragments > 1 mm in LEO.
NASA-STD-8719.14C Req 4.4-2	Accidental explosion: Design & plan to deplete, disconnect, or control stored energy to prevent qualifying debris-producing explosion.
NASA-STD-8719.14C Req 4.5-1	NASA project requirement: Probability < 0.001 for accidental collision with space objects > 10 cm over mission orbital lifetime.
NASA-STD-8719.14C Req 4.6-4	Post-mission disposal reliability: Probability of successful post-mission disposal ≥ 0.90 at end of mission.
NASA-STD-8719.14C Req 4.7-1	Human casualty risk: $E_c < 10^{-4}$ (1:10,000) for uncontrolled atmospheric entry; 15 J kinetic energy hazard threshold.
ISO 24113:2023	International voluntary consensus standard: Space systems debris mitigation requirements and conditional disposal targets.
ESA ESSB-ST-U-007 (2023)	ESA space debris mitigation requirements for ESA projects; distinct from the voluntary Zero Debris Charter 2030 pledge.
U.S. Gov. ODMSP (2019)	U.S. Government Orbital Debris Mitigation Standard Practices; non-binding federal baseline guidance.
FAA Part 450 (2023 NPRM)	Notice of Proposed Rulemaking (RIN 2120-AK81): Proposed launch stage disposal requirements; draft regulatory status.
ISRO DFSM (2024)	India's Intent on Debris-Free Space Missions — Explained: Institutional initiative and forward-looking sustainability pledge.
NIST AI 100-1 (AI RMF)	Trustworthy AI Risk Management Framework: Guidelines for explainability, data integrity, and human governance.

4 Related Work & Comparative Landscape

To contextualize OrbitReady's prototype boundary, Table 2 benchmarks the platform against existing agency software suites and commercial SSA platforms across functional capability dimensions.

Table 2: Functional comparison between OrbitReady, agency assessment software, and operational SSA platforms.

Functional Dimension	Agency / Commercial Baselines	OrbitReady Prototype
Primary Purpose	DAS/DRAMA: Agency compliance; SSA: Live tracking	Early-stage evidence pre-screening & gap identification
Target Operators	Agency contractors; well-funded prime constellations	SmallSat, NewSpace startups, university CubeSats
Interface Architecture	Compiled desktop GUI / legacy DLL executables	Native REST API, CLI, and web review console
Document Assistance	Manual parameter extraction from drawings	DocLLM layout-aware multimodal parameter extraction
Statutory Scope	Single agency jurisdiction (NASA-only or ESA-only)	Multi-jurisdiction source-linked rule fixtures
Trade Exploration	Manual iterative re-runs with user-selected inputs	Automated minimal-delta counterfactual parameter trades
Audit & Provenance	Local text output reports; proprietary SSA logs	W3C PROV-O DAG with SHA-256 tamper-evident chain
Operational Status	Operational agency standards / commercial platforms	Open-core pre-review prototype (Hackyard Build 2026)

5 OrbitReady System Architecture

To resolve the fundamental tension between flexible natural language document processing and rigorous rule evaluation, OrbitReady is structured around a decoupled, three-tier architecture. A fundamental architectural invariant of OrbitReady is that large language models (LLMs) are strictly prohibited from performing numerical orbit propagation, calculating physical lifetimes, or making compliance determinations. Instead, deterministic physical algorithms and versioned rule solvers execute all evaluations, while AI models are confined to layout-aware document assistance.

5.1 Layer 1: Ingestion & Mission Profiling

Layer 1 extracts structured mission parameters from operator-supplied PDF specification sheets, propulsion manuals, battery documentation, and CAD metadata using DocLLM (Wang et al., 2024) [15]. When available, connectors ingest operational Two-Line Element (TLE) sets and CCSDS Orbit Mean-Elements Messages (OMM) from Space-Track and CelesTrak to populate initial Keplerian state vectors [16, 17].

5.2 Layer 2: Deterministic Rule Evaluator & Decay Heuristics

Layer 2 computes orbital decay estimates using semi-analytic King-Hele drag formulations coupled with Jacchia-Roberts and NRLMSISE-00 atmospheric density profiles. Deterministic rule solvers evaluate mission parameters against versioned rule profiles derived from FCC 22-74 (5-year disposal), NASA-STD-8719.14C (passivation and casualty expectation), and ISO 24113:2023.

5.3 Layer 3: Explainable Intelligence & Human Review Gate

Layer 3 generates actionable remediation suggestions using minimal-delta counterfactual exploration (Wachter et al., 2018) [18], provides deterministic rule execution traces, logs calculation lineage using Buneman et al.'s (2001) data provenance theory within a W3C PROV-O DAG [21], and enforces a mandatory human review gate based on Parasuraman et al.'s (2000) Level 4 automation taxonomy [22].

6 Astrodynamics & Prioritization Formulation

The central prioritization heuristic generated by the engine is the OrbitScore Composite Mission Index, denoted $S_{\text{mission}} \in [0, 100]$. S_{mission} is formulated as a multi-attribute utility function aggregating four physical and statutory evidence sub-indices:

$$S_{\text{mission}} = w_{\text{pmd}} S_{\text{pmd}} + w_{\text{expl}} S_{\text{expl}} + w_{\text{coll}} S_{\text{coll}} + w_{\text{cas}} S_{\text{cas}} \tag{1}$$

Prioritization Scoping: The prototype uses explicit, user-visible weights (canonical defaults: $w_{\text{pmd}} = 0.35$, $w_{\text{expl}} = 0.25$, $w_{\text{coll}} = 0.20$, $w_{\text{cas}} = 0.20$) chosen as an engineering prioritization assumption. No claim is made that these weights represent a validated physical hazard model, scientific probability, or regulatory formula. To ensure transparency, OrbitReady reports four discrete evidence states: (1) *Verified / Consistent with Profile*, (2) *Conditional (Assumptions Apply)*, (3) *Needs Evidence*, and (4) *Critical Evidence Gap*. A composite score is never permitted to override a critical evidence gap.

6.1 Post-Mission Disposal Index (S_{pmd})

Evaluates whether the spacecraft clears LEO within configured statutory targets (5.0 yrs under FCC 22-74; 25.0 yrs under ISO 24113):

$$S_{\text{pmd}} = f_{\text{time}}(T_{\text{decay}}, T_{\text{stat}}) \cdot \left[\alpha_{\text{pmd}} P_{\text{pmd}} + (1 - \alpha_{\text{pmd}}) \min\left(1.0, \frac{\Delta V_{\text{avail}}}{\Delta V_{\text{req}}}\right) \right] \tag{2}$$

where P_{pmd} denotes modeled disposal reliability (target ≥ 0.90), and f_{time} penalizes decay times exceeding statutory limit T_{stat} piecewise.

6.2 Accidental Explosion & Passivation Evidence Index (S_{expl})

Evaluates stored chemical, kinetic, and electrical energy depletion evidence per NASA Req 4.4-2 and passivation literature [13, 14]:

$$S_{\text{expl}} = 100 \cdot \left(1 - \min\left(1.0, \frac{P_{\text{expl}}}{10^{-3}}\right)\right) \cdot [\beta_{\text{pass}} E_{\text{tank}} + (1 - \beta_{\text{pass}}) E_{\text{batt}}] \tag{3}$$

where $E_{\text{tank}} = 1.0$ if verified documentation shows a propellant venting or depletion plan that prevents qualifying debris-producing explosion or deflagration, and $E_{\text{batt}} = 1.0$ if battery cell discharge and isolation are evidenced.

6.3 Conjunction Screening Heuristic & Reentry Casualty Indices

OrbitReady implements an illustrative conjunction screening heuristic across collision cross-section A_{coll}:

$$S_{\text{coll}} = 100 \cdot \max\left(0.0,\ 1.0 - \frac{P_{\text{coll}}}{10^{-3}}\right) \quad [\text{Illustrative Heuristic}] \tag{4}$$

Operational Boundary Note: Operational conjunction assessment requires authoritative high-precision ephemerides with full covariance, screening conventions, encounter geometry, and operational tracking context that cannot be derived from public GP/TLE data. In OrbitReady's MVP, Equation (4) serves strictly as an illustrative screening heuristic; operational conjunction assessment is deferred to authoritative SSA platforms. Reentry casualty risk evaluates human safety limits for uncontrolled atmospheric reentry:

$$S_{\text{cas}} = 100 \cdot \max\left(0.0,\ 1.0 - \frac{E_c}{10^{-4}}\right) \tag{5}$$

evaluating whether uncontrolled reentry complies with the statutory human casualty expectation threshold $E_c < 10^{-4}$ (1:10,000 threshold), referencing NASA's 15 J kinetic energy threshold for potentially hazardous surviving fragments.

7 Explainable Intelligence & Counterfactual Optimization

OrbitReady formulates regulatory remediation suggestions through the mathematical framework of counterfactual explanations (Wachter, Mittelstadt & Russell, 2018) [18]. Given a mission parameter vector $\mathbf{x}_{\text{orig}}$ yielding an evidence gap $S_{\text{mission}}(\mathbf{x}_{\text{orig}}) < 85.0$, the engine solves for the minimal-delta parameter adjustment $\Delta\mathbf{x} = \mathbf{x}^* - \mathbf{x}_{\text{orig}}$ such that:

$$x^* = \arg\min_{x \in \mathcal{C}_{\text{feasible}}} \left[d(x, x_{\text{orig}}) + \lambda (85.0 - S_{\text{mission}}(x))^2 \right] \tag{6}$$

subject to physical spacecraft feasibility constraints $\mathcal{C}_{\text{feasible}}$ (propellant mass, dispenser envelope, power budgets).

Reproducible Counterfactual Sensitivity Trade: For a synthetic 12U SmallSat fixture with mass m = 21.0 kg, cross-sectional area A = 0.045 m^2, drag coefficient C_D = 2.2 (ballistic coefficient $B \approx$ 212.1 kg/m^2), deployed at 620 km circular SSO with mean solar flux $F_{10.7}$ = 120 sfu, the deterministic semi-analytic lifetime is 8.4 years (exceeding the FCC 5-year limit). The Counterfactual Engine computes two minimal-delta design options: (1) *Deployment Trade:* Lower insertion perigee by 38 km (to 582 km), reducing calculated lifetime to 4.2 years; OR (2) *Propulsion Trade:* Allocate 320 g of cold-gas propellant for an 18.4 m/s perigee-lowering burn to 490 km. Sensitivity analysis across solar flux cycles ($F_{10.7} \in$ [70, 220] sfu) shows that disposal lifetime varies from 3.1 to 6.2 years at 582 km, providing engineers with an explicit trade envelope.

7.1 Deterministic Rule Execution Traces

Rather than relying on opaque statistical attribution models (SHAP/LIME) that can obscure hand-authored legal thresholds, OrbitReady decomposes findings using exact deterministic rule traces:

$$\Delta S_{\text{mission}} = \sum_{j=1}^{M} \Delta S_j, \quad \Delta S_{\text{vent}} = +18.4 \text{ pts} \tag{7}$$

Each finding emits a structured trace: *Finding:* Needs Evidence; *Trigger:* Disposal lifetime = 8.4 yr; *Profile:* FCC 22-74 §25.283(e); *Threshold:* ≤ 5.0 yr; *Recommended Action:* Provide propulsion disposal budget or adjust insertion orbit.

8 Evidence Confidence & Cryptographic Provenance

In pre-review workflows, confidence must reflect the provenance and authoritativeness of underlying evidence rather than uncalibrated neural confidence. OrbitReady categorizes confidence across three provenance tiers: (1) *High Confidence:* Primary authoritative engineering document (CAD export, propulsion acceptance test report) matching serial numbers; (2) *Medium Confidence:* Secondary specification sheet or vendor brochure; (3) *Low Confidence:* Model extraction assumption or unresolved parameter conflict requiring human clarification.

Furthermore, based on Buneman, Khanna, and Tan's (2001) 'why and where' provenance theory [21], OrbitReady records every calculation step within a W3C PROV-O directed acyclic graph (DAG). Each node logs source document hashes, rule version IDs, and execution timestamps chained via cryptographic SHA-256 hashing:

$$H_{\text{node}} = \text{SHA256}(\text{Payload} \parallel H_{\text{parent}} \parallel \text{Timestamp}) \tag{8}$$

This architecture provides a tamper-evident audit trail demonstrating exactly which inputs and rule versions produced each finding in the exported review packet.

9 Human-in-the-Loop Decision Governance

OrbitReady establishes strict human-in-the-loop (HITL) boundaries based on the classic automation taxonomy of Parasuraman, Sheridan, and Wickens (2000) [22]. Table 3 details the four automation stages and OrbitReady's explicit boundaries.

Table 3: Levels of Automation (LOA) governance mapping across the four Parasuraman stages.

Stage & Function	Autonomy & Safety Boundary in OrbitReady
1. Info Acquisition	Level 8: High Automation; DocLLM document parsing with visual confidence highlighting.
2. Info Analysis	Level 6: Machine Execution; deterministic rule evaluation and lifetime heuristics.
3. Decision Selection	Level 4: System Suggests; OrbitReady proposes trade options; qualified human reviewer evaluates.
4. Action Implementation	Level 1: Strictly Manual; OrbitReady NEVER submits regulatory filings or executes commands autonomously.

Design Constraint Rationale: Level 4 is an intentional architectural design constraint, not an automated safety certification. By halting automated authority at recommendation, OrbitReady acts as a cognitive assistant that empowers qualified human reviewers while eliminating manual extraction fatigue.

10 Synthetic Case Studies & Experimental Demonstrations

We evaluated OrbitReady across three labeled synthetic satellite mission profiles to demonstrate deterministic rule behavior and evidence gap identification:

Synthetic Fixture 1: 3U Imaging CubeSat (Passive Deorbit). Parameters: mass m = 4.2 kg; cross section A = 0.035 m^2; 550 km circular SSO; no onboard propulsion. Rule Evaluation: Natural decay calculated at T_{decay} = 3.8 years (consistent with FCC 5-year post-mission window). Passivation plan confirmed via circuit depletion documentation. Reentry casualty analysis confirms small aluminum mass demises aerothermally. *Pre-review Status:* **No critical evidence gaps under configured profile**. Complete cited review packet compiled in 4.2 seconds.

Synthetic Fixture 2: 12U IoT SmallSat (Active Propulsion Trades). Parameters: dry mass m_{dry} = 18.5 kg, cold-gas propellant m_{fuel} = 2.5 kg at 180 bar; circular SSO at 700 km. Initial Evaluation: Unassisted decay is 38 years (violating FCC 5-year limit). Stored propellant lacks documented venting plan. *Pre-review Status:* **Critical evidence gaps identified**. Counterfactual Engine generates two remediation options: (1) Lower perigee to 320 km via 46.2 m/s burn (lifetime 1.4 years); (2) Evidence tank venting plan to deplete residual pressure. Post-remediation status: **Consistent with configured profile**.

Synthetic Fixture 3: Constellation Batch Profiling. 100 micro-satellites (120 kg each) deployed across two orbital planes at 650 km. Batch ingestion processes constellation ephemeris sets and identifies shared disposal risk factors. *Pre-review Status:* **Illustrative batch pre-screening completed; operational conjunction analysis deferred to authoritative SSA platforms**.

11 Hackyard Build 2026 Implementation Blueprint

OrbitReady was engineered and built by Team TwinForge (Narayan Kumar Jha, Aditya Kumar Sharma, Roshan Kumar Singh) specifically for the Hackyard Build 2026 24-hour physical build at IIT Guwahati (24–25 October 2026). Selected as a Wave 3 Finalist (Project ID: 4afc0da9-895d-4727-befa-772166bd5dd9; verified live at https://hackyard.org/build2026/result), the platform aligns directly with the official four-pillar Hackyard judging criteria in Table 4.

Table 4: Hackyard Build 2026 official four-pillar judging rubric alignment and system deliverables.

Official Criterion & Weight	Implementation Architecture & Verifiable Deliverable
Build (40%) ***Working software & execution***	End-to-end working prototype: FastAPI modular backend, React/Tailwind pre-review console, deterministic physical rule solver, and offline-first containerized architecture for the 24-hour sprint at IIT Guwahati.
Problem (20%) ***Meaningful, real-world challenge***	Addresses the acute operational bottleneck faced by emerging small-satellite operators: navigating fragmented, multi-jurisdictional debris regulations and multi-month manual ODAR preparation delays.
Ingenuity (20%) ***Clever & thoughtful approach***	Decoupled 3-layer architecture; strict isolation of AI from numerical physics; minimal-delta counterfactual sensitivity trade engine; W3C PROV-O tamper-evident calculation lineage.
Potential (20%) ***Longevity & future growth***	Open-core foundation with immediate utility for university CubeSat teams and NewSpace startups; clear post-hackathon roadmap to export official NASA DAS / ESA DRAMA file formats.

12 Conclusion & Future Outlook

As the commercial space sector accelerates toward tens of thousands of active satellites in Low Earth Orbit, fragmented and manual compliance evidence assembly represents an unsustainable operational bottleneck. In this paper, Team TwinForge presented OrbitReady, a source-linked pre-review copilot for spacecraft debris-mitigation evidence engineered for Hackyard Build 2026. By decoupling layout-aware document assistance from deterministic rule solvers and grounding trade studies in minimal-delta counterfactual analysis, transparent confidence tiers, and tamper-evident provenance logging, OrbitReady empowers small satellite teams to prepare rigorous evidence packets before formal expert submission.

Future enhancements following the Hackyard Build 2026 physical round will expand OrbitReady's propagation engine to incorporate high-order geopotential harmonics (J2–J6), integrate direct export into NASA DAS and ESA DRAMA native project files, and expand rule profiles to cover emerging national licensing frameworks.

Broader Impact & Ethical Considerations

This work democratizes access to orbital debris mitigation pre-screening for resource-constrained university teams, emerging startups, and developing nations, promoting space sustainability and orbital stewardship. To prevent potential over-reliance, OrbitReady enforces strict Human-in-the-Loop design constraints (Parasuraman Level 4 LOA), explicitly prohibiting autonomous filing submissions without qualified human engineering review.

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